

Introductory Survey on Temporal Hate Speech

Alberto Lorente Galé

October 14, 2025

Words and expressions may take on, or loose, hateful meanings due to changes in a language or societal norms (McGillivray et al., 2022). These changes may not be driven exclusively by linguistic processes, but also socio-political and cultural events which may alter people’s perception of each other and introduce new biases.

In current literature, the diachronic dimension of NLP systems is dealt with in two prominent ways. On one side there are time-agnostic, time-aware and temporal adaptations, which broadly aim to develop models that will perform adeptly in data from different ranges of time and make models more robust to future data. On the other side, there is lexical semantic change, which seeks to answer the question ”how does the meaning of a word change through time”.

When it comes to Hate Speech Detection, both are needed: it is obvious that if a model’s performance is too reliant on a particular time-specific event, its performance will not be transferred to other event-heavy hate speech (Jin et al., 2023). Parallely, overly relying on specific vocabulary may have catastrophic results since the words used to express a certain meaning can change and its use can decrease in favor of different verbalizations of expressing that same meaning (Lieberman et al., 2007). Even more challenging is the case where the ground truth the model is trying to predict changes, similar to the case of evolving legal doctrine (T.y.s.s et al., 2024).

1 Time-Agnosticism, Time-Awareness and Temporal Adaptations

Time-related research tends to fall under two categories: those works which propose time-specific solutions to the performance degradation (T.y.s.s et al. (2024); Yao et al. (2018)) and those which, although may propose more accessible mitigation strategies, rather focus on the evaluation of models trained with data from determined spans of time against test data from different spans, usually future data (Loureiro et al. (2022a); Florio et al. (2020); Röttger and Pierrehumbert (2021); Jin et al. (2023)).

Regarding the former trend, Yao et al. (2018) aimed to develop time-agnostic static embeddings by training individual time-specific embeddings and aligning them across time and T.y.s.s et al. (2024) tested incremental and time-invariant training paradigms with law-specific BERT variants (Chalkidis et al. (2020a) ; Chalkidis et al. (2020b)) for tasks related to the law domain. Overall, they found that time-invariant methods underperformed in their tasks and continual learning methods, namely LoRA (Hu et al., 2022) and Adapter (Houlsby et al., 2019) methods, excelled.

Regarding the latter trend, Loureiro et al. (2022a) continued pretraining of RoBERTa (Zhuang et al., 2021) time-specific models with data collected from Twitter between 2018 and 2021 and observed a clear degradation in performance when models trained with older data were tested with newer data.

As far as abusive language detection, Jin et al. (2023) assessed the temporal degradation of BERT (Devlin et al., 2019), RoBERTa (Zhuang et al., 2021) and RoBERTa-hate-speech (Vidgen et al., 2021) in English, Spanish, Chinese and Italian by comparing the performance of models trained in data split in training-testing sets randomly against models trained in chronological training-testing splits. Contrary to Loureiro et al. (2022a), Jin et al. (2023) only developed one model per dataset, instead of multiple time-defined ones, and their experimental setup resembled that of Online Continual Learning (Biesialska et al. (2020); Bidaki et al. (2025)); their data was sorted by ascending date and they separated the train, validation and test splits according to the date, taking the 70% of older data for the training split and the 15% that corresponded to the more recent data for the test split, leaving the rest for validation. The results also pointed to a degradation in the performance on recent data when

PLMs were trained on older data, an effect that was attributed to the appearance of events and topics in the testing data which were not part of the training data.

Röttger and Pierrehumbert (2021) arrived at a similar conclusion: exogenous events can drive changes in language and masking tokens in proper and common nouns in their upstream MLM temporal adaptation task drove a significant part of the improvement in their BERT model. The study also observed that while temporal adaptation improved the performance of the upstream task, it did not translate into an improvement in the downstream tasks with respect to general domain adaptation as it introduced extraneous event-specific information.

Lastly, Florio et al. (2020) tested the temporal robustness of AIBERTO (Polignano et al., 2019) when monitoring hate speech targeting immigrants in Twitter through the “Contro l’odio” platform. Overall, the study also underlined how event and topic-specific information degrades the performance of models when tested on future data and recommended training models with data as recent as possible.

Considering the aforementioned works, there are several identifiable challenges and limitations that current time-agnostic NLP research faces:

1. Overall, static methods *chase the future* because without further updates, they will obviously lag behind the present time.
2. The evaluation focuses mostly on the performance in future data. This fact does not pose a significant issue since authors point to the fact that models at a current time perform adequately on past data.
3. It is unanimously agreed that the further the test data is from the train data, the worse the performance is.
4. As far as data limitations, data is assumed to be already pre-split diachronically and available. However, where to split, where to conclude and where to start a time frame is not a trivial question.
5. How should event and topic information be dealt with? Some authors choose to mask or remove this kind of information, which can be problematic in HSD because emerging types of hate speech are driven by changing socio-political events although the “stream of events” identified by the previous works is the main cause of degradation (Florio et al. (2020); Röttger and Pierrehumbert (2021); Jin et al. (2023)). Intuitively speaking, in order for a HSD system to perform well, one could think that it should incorporate topic specific information but time-agnosticism may push models to favor generality. That is, making temporal-agnostic systems may have inherent limitations. If the performance focus is current and future data, adopting a continual learning paradigm could be beneficial, since new topics and events would be accounted for, although such a model would not be as generalizable. All in all, this may leave us at a crossroads between making time-agnostic systems that are too general or making continual temporal adaptations that may not be general enough and too specific to current events.

2 Lexical Semantic Change Detection

Lexical Semantic Change Detection (LSCD) aims to answer questions regarding the change of word meaning through time. There are two major formulations of LSCD. The first one sets out to study the change of particular words from one already delimited corpus to another corpus, which was produced further into the future, and the second one compares the meaning of a word in a pair of sentences (Words in Context).

Kiyama et al. (2025) proposed a framework for LSCD by computing a similarity matrix between embeddings of the same word in corpora of arbitrary time periods and clustering the similarity matrices for different words to categorize those ones which exhibit similar time shift behavior. The study also pointed out the limitation that a change in the similarity between two embeddings does not equate to a semantic shift.

Another well-known study in the field is Schlechtweg et al. (2020). They proposed a shared task consisting of detecting words that gained sense in a corpus with respect to a previous one and to rank the words according to the degree by which they acquired new meaning, both in an unsupervised way. The submissions were described as a combination of semantic representations, alignment techniques

and change measures and the authors noted that, to their surprise, static embeddings performed better than contextualized ones.

Ushio et al. (2022) studied quick temporal shifts in twitter for NER and constructed TweetNER7, a dataset collected by querying tweets with weekly trending keywords. In order to deal with the issue of short-term degradation, they proposed a self-labelling method, a sequential finetuning framework and a tweet contextualization technique that resembles current RAG (Lewis et al., 2020) approaches.

The only prominent example of LSCD in the Hate Speech domain, to our knowledge, is Hoeken et al. (2023), which measured lexical change of hate speech in historical data (16th century early modern English religious writings during the protestant reform) by projecting embeddings into a “hate” dimension.

As far as using LLMs, Wang and Choi (2023) showed how zero-shot prompting GPT-4 outperformed BERT and traditional statistical methods, such as Positive Pointwise Mutual Information, in TempoWiC (Loureiro et al., 2022b), and including the date of the tweet in the prompt resulted in a slight improvement in F1 score although the accuracy did not improve.

LSCD faces some similar challenges to temporal adaptation:

1. When determining if a word changed meanings between two corpora, the corpora is already split. In this case, we also need to determine which words would be compared.
2. When it comes to Words in Context, both pairs of texts and the word of interest are needed.
3. As some authors point out, a change in the distribution of the words with respect to their context does not mean that its meaning changed as well.
4. Most of the cited literature still reports that static embedding methods outperform contextual embeddings.

References

- Bidaki, S. A., Mohammadkhah, A., Rezaee, K., Hassani, F., Eskandari, S., Salahi, M., and Ghassemi, M. M. (2025). Online continual learning: A systematic literature review of approaches, challenges, and benchmarks.
- Biesialska, M., Biesialska, K., and Costa-jussà, M. R. (2020). Continual lifelong learning in natural language processing: A survey. In Scott, D., Bel, N., and Zong, C., editors, *Proceedings of the 28th International Conference on Computational Linguistics*, pages 6523–6541, Barcelona, Spain (Online). International Committee on Computational Linguistics.
- Chalkidis, I., Fergadiotis, M., Kotitsas, S., Malakasiotis, P., Aletras, N., and Androutsopoulos, I. (2020a). An empirical study on large-scale multi-label text classification including few and zero-shot labels. In Webber, B., Cohn, T., He, Y., and Liu, Y., editors, *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 7503–7515, Online. Association for Computational Linguistics.
- Chalkidis, I., Fergadiotis, M., Malakasiotis, P., Aletras, N., and Androutsopoulos, I. (2020b). LEGAL-BERT: The muppets straight out of law school. In Cohn, T., He, Y., and Liu, Y., editors, *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 2898–2904, Online. Association for Computational Linguistics.
- Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In Burstein, J., Doran, C., and Solorio, T., editors, *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.
- Florio, K., Basile, V., Polignano, M., Basile, P., and Patti, V. (2020). Time of your hate: The challenge of time in hate speech detection on social media. *Applied Sciences*, 10(12).

- Hoeken, S., Spliethoff, S., Schwandt, S., Zarri  , S., and Alacam,  . (2023). Towards detecting lexical change of hate speech in historical data. In Tahmasebi, N., Montariol, S., Dubossarsky, H., Kutuzov, A., Hengchen, S., Alfter, D., Periti, F., and Cassotti, P., editors, *Proceedings of the 4th Workshop on Computational Approaches to Historical Language Change*, pages 100–111, Singapore. Association for Computational Linguistics.
- Houlsby, N., Giurciu, A., Jastrzebski, S., Morrone, B., De Laroussilhe, Q., Gesmundo, A., Attariyan, M., and Gelly, S. (2019). Parameter-efficient transfer learning for NLP. In Chaudhuri, K. and Salakhutdinov, R., editors, *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2790–2799. PMLR.
- Hu, E. J., yelong shen, Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., and Chen, W. (2022). LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.
- Jin, M., Mu, Y., Maynard, D., and Bontcheva, K. (2023). Examining temporal bias in abusive language detection.
- Kiyama, H., Aida, T., Komachi, M., Ogiso, T., Takamura, H., and Mochihashi, D. (2025). Analyzing continuous semantic shifts with diachronic word similarity matrices. In Rambow, O., Wanner, L., Apidianaki, M., Al-Khalifa, H., Eugenio, B. D., and Schockaert, S., editors, *Proceedings of the 31st International Conference on Computational Linguistics*, pages 1613–1631, Abu Dhabi, UAE. Association for Computational Linguistics.
- Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., K  ttler, H., Lewis, M., Yih, W.-t., Rockt  schel, T., Riedel, S., and Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive nlp tasks. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474. Curran Associates, Inc.
- Lieberman, E., Michel, J.-B., Jackson, J., Tang, T., and Nowak, M. (2007). Quantifying the evolutionary dynamics of language. *Nature*, 449:713–6.
- Loureiro, D., Barbieri, F., Neves, L., Espinosa Anke, L., and Camacho-collados, J. (2022a). TimeLMs: Diachronic language models from Twitter. In Basile, V., Kozareva, Z., and Stajner, S., editors, *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics: System Demonstrations*, pages 251–260, Dublin, Ireland. Association for Computational Linguistics.
- Loureiro, D., D’Souza, A., Muhajab, A. N., White, I. A., Wong, G., Espinosa-Anke, L., Neves, L., Barbieri, F., and Camacho-Collados, J. (2022b). TempoWiC: An evaluation benchmark for detecting meaning shift in social media. In Calzolari, N., Huang, C.-R., Kim, H., Pustejovsky, J., Wanner, L., Choi, K.-S., Ryu, P.-M., Chen, H.-H., Donatelli, L., Ji, H., Kurohashi, S., Paggio, P., Xue, N., Kim, S., Hahm, Y., He, Z., Lee, T. K., Santus, E., Bond, F., and Na, S.-H., editors, *Proceedings of the 29th International Conference on Computational Linguistics*, pages 3353–3359, Gyeongju, Republic of Korea. International Committee on Computational Linguistics.
- McGillivray, B., Alahapperuma, M., Cook, J., Di Bonaventura, C., Mero  o-Pe  uela, A., Tyson, G., and Wilson, S. (2022). Leveraging time-dependent lexical features for offensive language detection. In Barbieri, F., Camacho-Collados, J., Dhingra, B., Espinosa-Anke, L., Gribovskaya, E., Lazaridou, A., Loureiro, D., and Neves, L., editors, *Proceedings of the First Workshop on Ever Evolving NLP (EvoNLP)*, pages 39–54, Abu Dhabi, United Arab Emirates (Hybrid). Association for Computational Linguistics.
- Polignano, M., Basile, P., Degemmis, M., Semeraro, G., and Basile, V. (2019). Alberto: Italian bert language understanding model for nlp challenging tasks based on tweets. In *Italian Conference on Computational Linguistics*.
- R  ttger, P. and Pierrehumbert, J. (2021). Temporal adaptation of BERT and performance on downstream document classification: Insights from social media. In Moens, M.-F., Huang, X., Specia, L., and Yih, S. W.-t., editors, *Findings of the Association for Computational Linguistics: EMNLP 2021*, pages 2400–2412, Punta Cana, Dominican Republic. Association for Computational Linguistics.

- Schlechtweg, D., McGillivray, B., Hengchen, S., Dubossarsky, H., and Tahmasebi, N. (2020). SemEval-2020 task 1: Unsupervised lexical semantic change detection. In Herbelot, A., Zhu, X., Palmer, A., Schneider, N., May, J., and Shutova, E., editors, *Proceedings of the Fourteenth Workshop on Semantic Evaluation*, pages 1–23, Barcelona (online). International Committee for Computational Linguistics.
- T.y.s.s, S., Vuong, T.-Q., and Grabmair, M. (2024). ChronosLex: Time-aware incremental training for temporal generalization of legal classification tasks. In Ku, L.-W., Martins, A., and Srikumar, V., editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3022–3039, Bangkok, Thailand. Association for Computational Linguistics.
- Ushio, A., Barbieri, F., Sousa, V., Neves, L., and Camacho-Collados, J. (2022). Named entity recognition in Twitter: A dataset and analysis on short-term temporal shifts. In He, Y., Ji, H., Li, S., Liu, Y., and Chang, C.-H., editors, *Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 309–319, Online only. Association for Computational Linguistics.
- Vidgen, B., Thrush, T., Waseem, Z., and Kiela, D. (2021). Learning from the worst: Dynamically generated datasets to improve online hate detection. pages 1667–1682.
- Wang, R. and Choi, M. (2023). Large language models on lexical semantic change detection: An evaluation.
- Yao, Z., Sun, Y., Ding, W., Rao, N., and Xiong, H. (2018). Dynamic word embeddings for evolving semantic discovery. In *Proceedings of the Eleventh ACM International Conference on Web Search and Data Mining*, WSDM ’18, page 673–681, New York, NY, USA. Association for Computing Machinery.
- Zhuang, L., Wayne, L., Ya, S., and Jun, Z. (2021). A robustly optimized BERT pre-training approach with post-training. In Li, S., Sun, M., Liu, Y., Wu, H., Liu, K., Che, W., He, S., and Rao, G., editors, *Proceedings of the 20th Chinese National Conference on Computational Linguistics*, pages 1218–1227, Huhhot, China. Chinese Information Processing Society of China.